

2ab

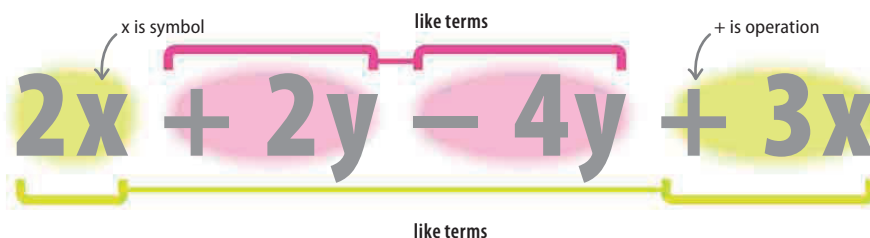
Working with expressions

AN EXPRESSION IS A COLLECTION OF SYMBOLS, SUCH AS X AND Y, AND OPERATIONS, SUCH AS + AND -. IT CAN ALSO CONTAIN NUMBERS.

Expressions are important and occur everywhere in mathematics. They can be simplified to as few parts as possible, making them easier to understand.

Like terms in an expression

Each part of an expression is called a "term." A term can be a number, a symbol, or a number with a symbol. Terms with the same symbols are "like terms" and it is possible to combine them.



Identifying like terms

The terms $2x$ and $3x$ are like terms because they both contain the symbol x . Terms $2y$ and $-4y$ are also like terms because each contains the symbol y .

Simplifying expressions involving addition and subtraction

When an expression is made up of a number of terms that are to be added or subtracted, there are a number of important steps that need to be followed in order to simplify it.

Write down the expression

Before simplifying the expression, write it out in a line from left to right.

$$3a - 5b + 6b - 2a + 3b - 7b$$

Group the like terms

Then group the like terms together, keeping the operations as they are.

$$3a - 2a - 5b + 6b + 3b - 7b$$

like terms like terms

Work out the result

The next step is to work out the result of each like term.

$$3a - 2a = 1a \quad \rightarrow \quad 1a - 3b \quad \leftarrow \quad -5b + 6b + 3b - 7b = -3b$$

Simplify the result

Further simplify the result by removing any 1s in front of symbols.

$$\text{term } 1a \text{ always written as } a \rightarrow a - 3b$$